



Session 11

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Module 2, Week 5
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Module 2: The Data Handling Toolkit: NumPy and Pandas



Identifying and treating missing values;

Data Cleaning in Data Processing Workflow;

Executing data format conversion techniques

What is Data Cleaning



- In real life, data is not as we want
- It contains Missing values, Incorrect data types, Duplicate rows, Inconsistent formatting, Extra spaces or unwanted symbols, Wrong or unrealistic values - *Data cleaning is the process of identifying and correcting*
- Pandas, the python library helps to clean the data

Name	Age	Salary
Ravi	25	50000
Anu	NaN	45000
ravi	25	50000
Priya	twenty-eight	60000

Data Cleaning: Import Pandas



- First step is to import Pandas
- Store or create dataframe
- Get to know the functions available in Pandas for data cleaning

Various types of undesirable states in data - Messy Data



1. Missing Values

Student	Marks
A	85
B	
C	90

2. Wrong Formats

Date written as 12/05/2026

Another date written as May 12, 2026

Age entered as "twenty" instead of 20

3. Duplicate Looking Values

Delhi
Delhi
DELHI

4. Extra Spaces, random symbols in place of text

Managing Missing Values



No Fixed Rule, some hints

1. If very few rows are missing, sometimes we can drop them.
2. For numerical columns like age or salary, we can fill with mean or median.
3. For text/category columns like department or city, we can fill with mode or Unknown.

Data Analysis Workflow and Data Cleaning

Data Collection

- Gather data from source (CSV files, databases, surveys, APIs, or sensors)

Data Cleaning

- Handle missing values
- Remove duplicates
- Correct formats
- Normalize data

Data Exploration and Analysis

- Compute statistics (Mean, Median, Quartiles, SD, etc)
- Identify trends and patterns (Get a feel of data distribution)

Visualization

- Create charts and graphs (to see the distribution in one page)

Machine Learning or Reporting

- Train ML algorithm, deliver the model for prediction

Pandas Data Cleaning Functions accessible using data frame



Function		Purpose
<code>df.isnull()</code>	-	Detect missing values
<code>df.dropna()</code>	-	Remove missing values
<code>df.fillna()</code>	-	Fill missing values
<code>df.duplicated()</code>	-	Detect duplicate rows
<code>df.drop_duplicates()</code>	-	Remove duplicates
<code>df.astype()</code>	-	Change data type
<code>df.replace()</code>	-	Replace incorrect values

Pandas Data Cleaning Functions accessible using Pandas



Function

Purpose

`pd.to_datetime()`

- Convert date into a fixed format

`pd.to_numeric()`-

Convert a string column into numeric

Why is Data Cleaning Important?

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- More accurate analysis of Attributes (Eg., How many people are from Delhi?)
 - Focused ML algorithm's training, no deviation due to data inconsistency (Eg., Duplicate entries/empty cells may mislead the algorithm)
 - Better performance on end-task
 - Reliable product/model



Thank you!